

HUDA KASIMOĞLU



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SUMMARY

Freshly graduated Software Engineer with hands-on experience in autonomous robotics and embedded systems. Developed autonomous ground and aerial vehicles using ROS 2, PX4, and real-time sensor integration. Passionate about robotics, drones, and embedded systems, seeking opportunities to grow in these fields.

EXPERIENCE

DOF Robotics, Long-Term Internship

Feb - Jun 2026

- Built autonomous ground and aerial vehicles using ROS 2, PX4, OpenCV, SLAM, and Nav2 — from simulation to real hardware with LiDAR, encoders, and PID control
- Developed a multi-threaded client-server chat application with file transfer using Python sockets and Qt

Piller, Summer Internship

Aug - Sep 2025

- Developed and prototyped access control and security systems using C/C++ within embedded environments.
- Designed access authorization algorithms by successfully integrating microcontrollers with various sensors and actuators.

EDUCATION

Bachelor of Software Engineering

Sep 2022- Jun 2026

Istanbul Sabahettin Zaim University

GPA: 3.87 - Ranked 3rd in Department and Faculty

Minor of Electrical & Electronics Engineering

Sep 2023 - Jun 2026

Istanbul Sabahettin Zaim University

GPA: 3.57

PROJECTS

Autonomous Fire Detection & Extinguishing UAV (Graduation Project)

- Built and flight-tested an autonomous UAV that detects smoke using a YOLOv8 model, calculates target GPS coordinates via vision-based geo-location, and navigates autonomously to the fire source
- Designed the full hardware and software stack: Pixhawk flight controller, Raspberry Pi 4B, ArduPilot, PyMAVLink, OpenCV, and UDP video streaming — validated first in Gazebo SITL, then in real flight tests with PID tuning via Mission Planner

Autonomous Differential-Drive Robot

- Built low-level motor control on ESP32 (discrete PID, encoder-based odometry, dual-core RTOS) bridged to ROS2 Humble on Raspberry Pi 4 via micro-ROS.
- Integrated URDF/TF, RPLIDAR C1, SLAM, and Nav2 for autonomous navigation; tested via teleop and live RViz visualization.

SKILLS

- Languages:** Python, C, C++
- Robotics & Embedded:** ROS2 Humble, ArduPilot, PX4, SLAM, Nav2, Gazebo, gz-sim
- Tools:** Git, Linux, Qt, Docker

LANGUAGES

- Arabic:** Native
- Turkish:** Native
- English:** Professional working proficiency
- YÖKDİL 72